

Print addresses of Code, Global, Static, Heap and Stack variables

```
saireddy@DESKTOP-HKQRL9G:~$ mkdir Memory_Practical
saireddy@DESKTOP-HKQRL9G:~$ cd Memory_Practical
saireddy@DESKTOP-HKQRL9G:~/Memory_Practical$ pwd
/home/saireddy/Memory_Practical
saireddy@DESKTOP-HKQRL9G:~/Memory_Practical$
```

```
GNU nano 7.2 memory_addresses.c *
#include <stdio.h>
#include <stdlib.h>

int global_var = 10;
static int static_var = 20;

void code_function()
{
    printf("This is a code function.\n");
}

int main()
{
    int stack_var = 30;
    int *heap_var = (int *)malloc(sizeof(int));

    *heap_var = 40;

    printf("Memory Address Information\n");
    printf("-----\n");

    printf("Code address   : %p\n", (void *)code_function);
    printf("Global address  : %p\n", (void *)&global_var);
    printf("Static address   : %p\n", (void *)&static_var);
    printf("Heap address     : %p\n", (void *)heap_var);
    printf("Stack address    : %p\n", (void *)&stack_var);

    free(heap_var);

    return 0;
}
```

```
/home/saireddy/Memory_Practical
saireddy@DESKTOP-HKQRL9G:~/Memory_Practical$ nano memory_addresses.c
saireddy@DESKTOP-HKQRL9G:~/Memory_Practical$ gcc memory_addresses.c -o memory_addresses
saireddy@DESKTOP-HKQRL9G:~/Memory_Practical$ ./memory_addresses
Memory Address Information
-----
Code address   : 0x5ec99045d1c9
Global address  : 0x5ec990460010
Static address   : 0x5ec990460014
Heap address     : 0x5ec9c2c352a0
Stack address    : 0x7ffc9f3c571c
saireddy@DESKTOP-HKQRL9G:~/Memory_Practical$
```

High Address

```
+-----+
| Stack   |
| stack_var |
+-----+
|         |
```

```

|   Heap   |
| heap_var |
+-----+
| Global / Static |
| global_var  |
| static_var  |
+-----+
|   Code   |
| code_function |
+-----+

```

Low Address

PRACTICAL 2

Study process memory using /proc/<pid>/maps

```

sairedy@DESKTOP-HKQRL9G:~/Memory_Practical$ ~/Memory_Practical
-bash: /home/sairedy/Memory_Practical: Is a directory
sairedy@DESKTOP-HKQRL9G:~/Memory_Practical$ Z|

```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
int global_var = 100;
```

```
static int static_var = 200;
```

```
int main()
```

```
{
```

```
    int stack_var = 300;
```

```
    int *heap_var = malloc(sizeof(int));
```

```
    *heap_var = 400;
```

```
    printf("Memory Mapping Program\n");
```

```
    printf("-----\n");
```

```
printf("PID      : %d\n", getpid());

printf("Global   : %p\n", (void *)&global_var);

printf("Static    : %p\n", (void *)&static_var);

printf("Heap      : %p\n", (void *)&heap_var);

printf("Stack     : %p\n", (void *)&stack_var);


printf("\nProcess is running for 60 seconds...\n");

sleep(60);

free(heap_var);

return 0;

}
```

Run the program in background